

# Choupo

The Open Source Chemical Process Simulator

## Properties Guide

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Authors: Vítor Geraldês, Pedro Mendes, and Miguel Rodrigues.

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*“Thermodynamics is a funny subject. The first time you go through it, you don’t understand it at all. The second time you go through it, you think you understand it, except for one or two points. The third time you go through it, you know you don’t understand it, but by that time you are so used to it, it doesn’t bother you any more.”*

— Arnold Sommerfeld

# 1 Introduction: the difficulty, the map, the promise

*What this guide is — and is not.* This is the props-**mode workflow** guide: how you *estimate* a missing compound, *fit* its binary pairs, and *consistency-test* the data — the see-then-decide loop of the Props/Explore workspace. It is *not* the property *theory*. For the first-principles derivations of NRTL, Wilson, UNIQUAC, UNIFAC, the cubic equations of state (SRK, Peng–Robinson), the electrolyte models (Pitzer, eNRTL, aqueous speciation), Henry’s law, Rackett, Ambrose–Walton, sublimation and the NASA–CEA Gibbs-of-formation data, see the **Theory Guide** (the in-GUI model panels deep-link straight into it).

If thermodynamics felt slippery the first time — the formalism pristine, yet somehow never self-sufficient when you sat down with real data — you are in good company: Sommerfeld, who reshaped quantum physics, said he never wrote his thermodynamics book because he never felt he had understood the subject. The difficulty is real, but it is not formal incompleteness. It is that an *axiomatic-looking* macroscopic theory rests, underneath, on emergent statistical facts and on parameters that must be *measured*, not derived. The confusion is the seam between the two showing through. This guide does not hide that seam — it makes it the organising idea.

## 1.1 The target: where theory grips

Picture property prediction as a target (Figure 1): pristine at the core, increasingly empirical toward the rim.

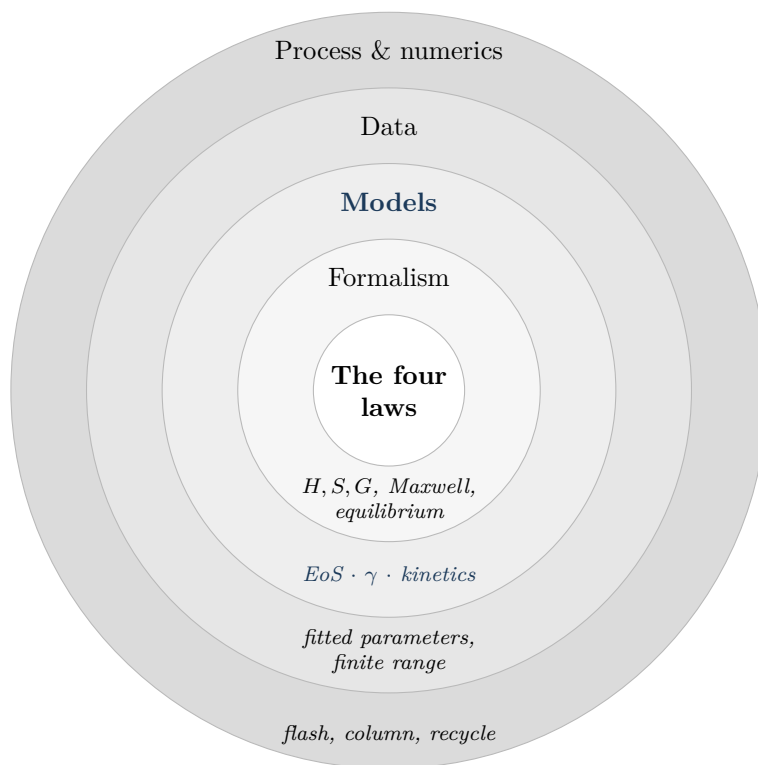


Figure 1: The layers of property prediction. The two inner shells follow from the laws alone. The accent shell — *constitutive models* — is the first fault line: the laws say *that* fugacities must match across phases, never *how* to compute one for a real mixture. From there outward, the theory is necessary but no longer sufficient.

- **The four laws (core).** The zeroth, first, second and third laws define temperature, internal energy, entropy, and — via Nernst — an absolute zero of entropy. Parameter-free.
- **Formalism.** Pure manipulation yields the potentials  $H$ ,  $G$ ,  $A$ , the Maxwell relations, the equilibrium criteria (equality of fugacity across phases,  $\Delta G = -RT \ln K$  for reaction) and the phase rule. Still derivable from nothing but the laws.
- **Constitutive models — where it first grips.** The laws require fugacities to be equal across phases; they do not tell you how to *compute* a fugacity for a real mixture. You must supply a model they do not give you — an equation of state, an activity model, a kinetic law. In Choupo: `idealGas/SRK/PR`; `ideal/NRTL/Wilson/UNIFAC`.
- **Data — “go to the laboratory.”** Those models carry parameters thermodynamics cannot predict:  $T_c$ ,  $P_c$ ,  $\omega$ , the Antoine constants,  $C_p^\circ(T)$ ,  $\Delta H_f^\circ$ ,  $S^\circ$ , the NRTL  $a_{ij}$ . Each is measured or estimated, and each has a finite

validity range. In Choupo, these are the `.dat` files.

- **Process & numerics.** Where the predicted numbers are consumed — flash, distillation, recycle — and grip a second time, now numerically (multiple EoS roots, phase stability, initialisation).

**Key idea.** Sommerfeld’s difficulty is this picture: a theory that *looks* axiomatic to the rim, but stops being self-sufficient at the accent shell. The honest move is not to paper over the seam between rigorous derivation and fitted correlation — it is to show exactly where it runs.

## 1.2 The glass-box promise

Because the outer shells are empirical, every number Choupo predicts should travel with two things attached:

- **Provenance** — which method produced it (rigorous derivation, fitted correlation, or group-contribution estimate) and the source of each parameter (a measured datum or an estimate).
- **Validity awareness** — whether a correlation is *extrapolating beyond its fitted range*, with a warning when it is, rather than a confident number where none is warranted.

*Intuition.* Ask Choupo for the vapour pressure of nitrogen at room temperature and it *refuses*:  $\text{N}_2$  is supercritical there (its critical temperature is  $-147^\circ\text{C}$ ), and a saturation pressure does not exist above the critical point. The  $P^{\text{sat}}$  curve simply *breaks* at  $T_c$  instead of extrapolating Antoine into a meaningless  $\sim 600$  bar. That refusal is the whole promise in miniature: the model declines to invent a number the physics forbids.

## 1.3 Where we are heading

A single property over a wide temperature span is not one problem — it is several physical regimes stitched together, and your confidence in any predicted number rises and collapses as you cross them. Figure 2 is the destination of this guide: the vapour pressure of a **water / ethanol / benzene** mixture from  $-200$  to  $1000^\circ\text{C}$ , drawn not as a curve but as a band of *confidence*.

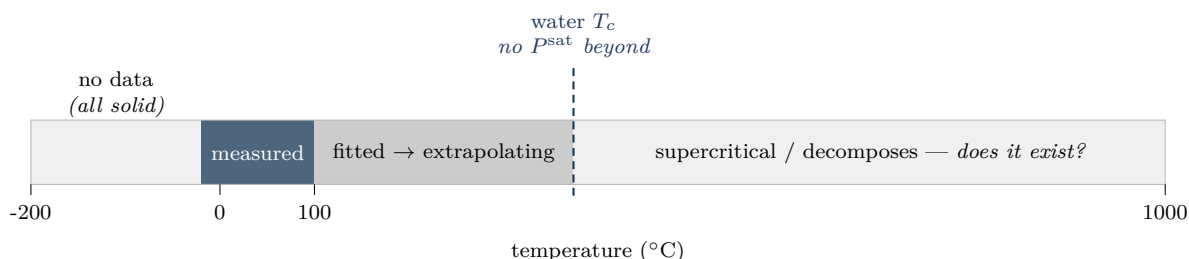


Figure 2: The same property, five honesties. A narrow window where we actually have data; degradation outward as the correlation extrapolates; a hard *wall* at the critical temperature where the property ceases to exist (the nitrogen refusal, generalised); and a cryogenic desert where every species has frozen solid (ethanol last, at  $-114^\circ\text{C}$ ) and there is simply no data. The theory runs to both edges; reality and data do not.

We build each piece of that picture in the chapters that follow — estimating a missing component (§6), curating the mixture models against lab points (§7), and testing whether the data itself is even consistent (§8). We read that span regime by regime at the end of this introduction (§1.6), and return to solve it numerically, end to end, at the close.

## 1.4 Anatomy of a component: where the numbers come from

Open any `.dat` and you see a flat list of numbers. The natural question — the one this whole guide turns on — is: *which of these are independent inputs, and which derived properties does the engine actually build from them?* Let us read `acetone.dat` field by field and follow each number to the property it feeds and to the line of code that consumes it.

| Field                                            | What it is, and what it builds                                                                                                            |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Identity</i>                                  |                                                                                                                                           |
| MW                                               | molar mass; converts molar $\leftrightarrow$ mass everywhere.                                                                             |
| <i>Caloric anchors — standardThermochemistry</i> |                                                                                                                                           |
| dHf_298 $\Delta H_f^\circ$                       | the <i>position</i> of the enthalpy scale: bond energy $C_p$ knows nothing about (combustion calorimetry).                                |
| s_298 $S^\circ$                                  | the <i>absolute</i> (third-law) entropy at 298 K — a true zero-referenced value, not a difference.                                        |
| <i>Caloric shape — idealGasHeatCapacity</i>      |                                                                                                                                           |
| coefficients $C_p^\circ(T)$                      | the master correlation: integrated to give the <i>temperature shape</i> of $H$ , $S$ , $G$ (below).                                       |
| <i>Corresponding-states trio</i>                 |                                                                                                                                           |
| Tc, Pc                                           | critical point: set the <i>size</i> of the cubic-EoS attraction term $a_c$ , and the reduced temperature $T_r = T/T_c$ .                  |
| omega $\omega$                                   | acentric factor: sets the <i>temperature shape</i> of that attraction, $\alpha(T_r; \omega)$ — how non-spherical / polar the molecule is. |
| <i>Phase change</i>                              |                                                                                                                                           |
| Tb, HvapTb                                       | boiling point + latent heat at $T_b$ ; with Tc drive Watson's $\Delta H_{\text{vap}}(T)$ .                                                |
| Vliq                                             | liquid molar volume (Poynting, liquid density).                                                                                           |
| <i>Correlations &amp; recipes</i>                |                                                                                                                                           |
| vaporPressure                                    | Antoine $\log_{10} P^{\text{sat}} = A - B/(T + C)$ , with its <b>Trange</b> ; the saturation curve.                                       |
| groups                                           | the molecular decomposition a group-contribution estimator / UNIFAC reads (the curation recipe, §6).                                      |
| diffusionVolume                                  | Fuller gas diffusivity $\Rightarrow$ Lewis number (wet-bulb).                                                                             |

## The caloric engine: $H$ , $S$ , $G$ from $C_p$

This is the mechanism behind “other properties that derive from here” — and it is worth seeing *how one arrives at*  $S(T)$ , since that is the step that trips everyone up. Start from the laws, at constant pressure. Heat capacity *is* the temperature slope of enthalpy,  $C_p = (\partial H / \partial T)_P$ , so

$$dH = C_p dT.$$

For entropy, the second law gives  $dS = \delta q_{\text{rev}}/T$ ; at constant pressure the reversible heat is  $\delta q_{\text{rev}} = C_p dT$ , hence

$$dS = \frac{C_p}{T} dT.$$

That single step — dividing by  $T$  — is the whole origin of the  $\int C_p/T$ . Integrate each from a reference temperature and add the 298 K anchor; Choupo stores  $C_p^\circ(T)$  and the two anchors (*not* tables of  $H$ ,  $S$ ) and evaluates:

$$H^\circ(T) = \underbrace{\Delta H_f^\circ}_{\text{position}} + \int_{298.15}^T C_p^\circ(T') dT' \quad (\text{Component}::\text{h\_pure\_ig}, \text{Component.cpp}) \quad (1)$$

$$S^\circ(T) = \underbrace{S_{298}^\circ}_{\text{absolute}} + \int_{298.15}^T \frac{C_p^\circ(T')}{T'} dT' \quad (\text{Component}::\text{s\_pure\_ig}) \quad (2)$$

$$G^\circ(T) = H^\circ(T) - T S^\circ(T) \quad (3)$$

For a mixture the engine adds the entropy of mixing  $-R \sum y_i \ln y_i$  and the pressure term  $-R \ln(P/P_{\text{ref}})$  (**ThermoPackage::S\_ig**). That is the *entire* ideal-gas caloric model — no hidden tables.

These integrals are not done numerically. For a polynomial  $C_p$  the primitive is elementary — it was solved *once, on paper* — so each call evaluates a closed form,

$$\int_{T_{\text{ref}}}^T \frac{C_p^\circ}{T'} dT' = a_0 \ln \frac{T}{T_{\text{ref}}} + \sum_{k \geq 1} \frac{a_k}{k} (T^k - T_{\text{ref}}^k),$$

exact to machine precision in a handful of operations (**PolynomialCp::S**). The standard correlations all share this property (NASA, Aly–Lee, Shomate have closed primitives too); Choupo never runs a quadrature loop for  $C_p$ .

**Worked example 1.1.** Acetone  $H$  and  $S$  at 400 K — by hand, then by ChoupoWith  $C_p^\circ(T) = 6.301 + 0.2606 T - 1.253 \times 10^{-4} T^2 + 2.038 \times 10^{-8} T^3$  (units  $\text{J mol}^{-1} \text{K}^{-1}$ ), integrating from 298.15 to 400 K:

$$\int C_p^\circ dT' = 8431 \text{ J/mol}, \quad \int \frac{C_p^\circ}{T'} dT' = 24.19 \text{ J/(mol} \cdot \text{K)}.$$

So  $H^\circ(400) = -217100 + 8431 = -208669$  J/mol and  $S^\circ(400) = 295.30 + 24.19 = 319.49$  J/(mol · K). Running `choupoProps` on a one-point scan returns exactly  $H_{\text{ig}} = -208669$  J/mol,  $S_{\text{ig}} = 319.49$  J/(mol · K). The number on the page is the number in the code — and at 298.15 K both integrals vanish, so  $H = \Delta H_f^\circ$  and  $S = S_{298}^\circ$ , the anchors laid bare.

**Key idea.**  $C_p$  gives the *shape*; the formation values give the *position*. That is why the `.dat` lists  $C_p^\circ(T)$  and  $\Delta H_f^\circ$ ,  $S^\circ$  separately — not redundancy. And note the asymmetry: entropy has an absolute zero for free (third law:  $S \rightarrow 0$  for a perfect crystal at  $T=0$ ), so  $S_{298}^\circ$  is a genuine absolute; enthalpy has *no* natural zero, so  $\Delta H_f^\circ$  must be *measured* and supplied.

## What the critical properties and acentric factor actually do

$T_c$ ,  $P_c$ ,  $\omega$  do nothing in the caloric equations above — they enter only when you ask for *real-fluid* behaviour through a cubic equation of state. There the attraction term is  $a(T) = a_c \alpha(T_r)$ , with  $a_c$  fixed by  $T_c$ ,  $P_c$  and the temperature shape fixed by  $\omega$  (Soave/SRK):

$$\alpha(T) = [1 + m(1 - \sqrt{T_r})]^2, \quad m = 0.48508 + 1.55171\omega - 0.15613\omega^2, \quad T_r = \frac{T}{T_c}$$

(`SRK.cpp`; Peng–Robinson uses the analogous  $\kappa(\omega)$ ). This  $a(T)$  is what produces the departure functions  $H - H^\circ$ ,  $S - S^\circ$  and the compressibility  $Z$  — the real-gas correction layered on top of the ideal-gas caloric result. The same  $T_c, T_b$  also drive Watson’s latent heat,  $\Delta H_{\text{vap}}(T) = \Delta H_{\text{vap}}(T_b) [(T_c - T)/(T_c - T_b)]^{0.38}$  (`Component.cpp`).

**Intuition.** So why does `acetone.dat` carry  $\omega$  at all if you are running the default `idealGas`? Because the data is *model-agnostic* and the model decides which fields wake up. Under `idealGas` there is no departure, and  $\omega, T_c, P_c$  are simply never read. Select `SRK` for a high-pressure compressor and the very same numbers come alive. The `.dat` is the laboratory record; the model is the question you ask of it.

## 1.5 From $G$ to equilibrium: what $\Delta G^\circ$ is (and is not)

The single most common confusion in this whole subject is  $\Delta G^\circ$  versus  $\Delta G$ . It is worth settling once, because  $G$  is what the reactors minimise and it is built from exactly the  $H$  and  $S$  above.

**First, the  $^\circ$ .** It means *reference state* — each species pure, ideal gas, at 1 bar — but *at the reaction temperature*  $T$ . It does *not* mean “at 298 K.” So  $\Delta G^\circ = \Delta G^\circ(T)$  varies with temperature;  $\Delta G^\circ(298)$  is just one point on a curve.

**Then, the two are not the same thing.**

- $\Delta G^\circ$  is the Gibbs change if reactants  $\rightarrow$  products with *everything in its standard state*. It is a *fixed number* — a property of the reaction at  $T$ .
- $\Delta G$  is the *actual* Gibbs change at the real composition (the real partial pressures). It is the *driving force*.

The reaction quotient  $Q$  bridges them, and equilibrium is where the driving force — not the reference number — vanishes:

$$\Delta G = \Delta G^\circ + RT \ln Q, \quad \Delta G = 0 \text{ at equilibrium} \Rightarrow \boxed{\Delta G^\circ = -RT \ln K}$$

**Key idea.** At equilibrium it is  $\Delta G$  that is zero, *never*  $\Delta G^\circ$ .  $\Delta G^\circ$  stays fixed and merely sets *where* the equilibrium sits (the value of  $K$ ); the system gets there by shifting  $Q$  until  $\Delta G = 0$ . And  $\Delta G^\circ > 0$  does not mean “no reaction” — it means  $K < 1$ , i.e. partial conversion. Only the sign and size of  $K$  change.

**Where the number comes from** — straight back to the caloric engine:

$$\Delta G^\circ(T) = \Delta H^\circ(T) - T \Delta S^\circ(T) = \sum_i \nu_i g_i^\circ(T), \quad g_i^\circ(T) = h_i^{\text{ig}}(T) - T s_i^{\text{ig}}(T).$$

Note the subtlety that confuses everyone:  $h^{\text{ig}}$  carries the *formation* enthalpy ( $\Delta H_f^\circ$ , an elements datum) while  $s^{\text{ig}}$  is the *absolute* (third-law) entropy. Mixing the two references looks wrong — but it is exactly right, because in a *balanced* reaction the element contributions cancel through  $\sum_i \nu_i$ . That is precisely why the `.dat` carries  $\Delta H_f^\circ$  and  $S^\circ$  separately: each plays a different role, and they combine correctly only across a balanced stoichiometry (`Component::g_pure_ig`; consumed in `PFR.cpp`/`CSTR.cpp` as  $K = \exp(-\Delta G^\circ/RT)$  and in the Gibbs reactor’s direct  $G$ -minimisation).

**How  $K$  moves with  $T$  (Gibbs–Helmholtz).**  $\Delta G^\circ(T)$  is not a frozen datum — it has its own temperature law, and that law is a workhorse relation worth knowing by name. Gibbs–Helmholtz states  $[\partial(G/T)/\partial T]_P = -H/T^2$ ; applied to  $\Delta G^\circ = -RT \ln K$  it collapses to the van 't Hoff equation,

$$\frac{d \ln K}{dT} = \frac{\Delta H^\circ}{RT^2}$$

which is just Le Chatelier made quantitative: an exothermic reaction ( $\Delta H^\circ < 0$ ) has  $K$  *falling* as  $T$  rises. In Choupo it doubles as a glass-box self-check — the  $K(T)$  the reactors build from  $g^\circ(T)$  at each temperature must show exactly this slope, or the enthalpy and the equilibrium constant are disagreeing with each other.

**Intuition.** These relations are not ornaments. Gibbs–Helmholtz is *how the equilibrium moves with temperature* (above); its sibling, the **Gibbs–Duhem** equation  $\sum_i x_i d \ln \gamma_i = 0$  (constant  $T, P$ ), says the activity coefficients of a mixture are *not independent*. That single constraint is exactly why measured VLE data can be *tested* for thermodynamic consistency — and why an activity model is built from one  $G^\text{E}$  function rather than fitting each  $\gamma_i$  freely. It is the engine of §8.

## 1.6 A worked span: water / ethanol / benzene, $-200$ to $1000^\circ\text{C}$

Put the three together and sweep the temperature. This one mixture crosses *every* regime of property prediction, and is the honest way to see both how properties are obtained *and* where the models give out — which is exactly the colour band of Figure 2, now read regime by regime. (Critical temperatures: ethanol  $241^\circ\text{C}$ , benzene  $289^\circ\text{C}$ , water  $374^\circ\text{C}$ ; melting points: ethanol  $-114$ , water  $0$ , benzene  $5.5^\circ\text{C}$ .)

| Regime                                                                       | How a property is obtained                                                                                                                             | Where the model gives out                                                                                                                                                                                  |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $-200$ to $\sim 0^\circ\text{C}$<br>solids (SLE)                             | melting points $T_m$ , $\Delta H_{\text{fus}}$ , solid $C_p$ ;<br>freezing-point depression; sublimation $P^{\text{sat}}$                              | <b>low-T edge.</b> Below every Antoine <b>Trange</b> : $P^{\text{sat}}$ would extrapolate blindly. Solid-phase data is sparse — a cryogenic <i>data desert</i> .                                           |
| $\sim 0$ to $\sim 65^\circ\text{C}$<br>two liquids (LLE)                     | activity model (NRTL/UNIFAC) for $\gamma$ ;<br>water/benzene nearly immiscible, ethanol the cosolvent                                                  | VLE-fitted parameters do <i>not</i> serve LLE; <b>Wilson</b> cannot represent a liquid split at all; without $T$ -dependent parameters the miscibility gap appears/vanishes wrongly.                       |
| $\sim 65$ to $\sim 100^\circ\text{C}$<br>boiling (VLLE)                      | Antoine $P^{\text{sat}}$ + $\gamma$ - $\phi$ + three-phase flash (Gibbs minimisation); ethanol/water + ternary azeotrope ( $\sim 64.9^\circ\text{C}$ ) | azeotrope sensitivity — a small $\gamma$ error moves the column design; bubble-point (Wang–Henke) distillation is unstable <i>through</i> an azeotrope (use <b>method simultaneous</b> ).                  |
| $\sim 100$ to $\sim 290^\circ\text{C}$<br>vapour $\rightarrow$ near-critical | cubic EoS (SRK/PR) with $k_{ij}$ ; departure functions $H - H^\circ$ , $S - S^\circ$ , $Z$                                                             | cubics degrade as the critical region is approached; $k_{ij}$ often needs to be made temperature-dependent over so wide a span.                                                                            |
| $\sim 290$ to $374^\circ\text{C}$<br>mixed super/sub-critical                | the method itself must switch $\gamma$ - $\phi \rightarrow \phi$ - $\phi$ (one EoS for every phase)                                                    | above a species' $T_c$ the activity-coefficient picture loses meaning; no single $\gamma$ model spans the switch; near-critical accuracy is poor.                                                          |
| $374$ to $1000^\circ\text{C}$<br>supercritical + re-action                   | $\phi$ - $\phi$ EoS; wide-range $C_p^\circ(T)$ for the caloric terms                                                                                   | <b>high-T edge.</b> $P^{\text{sat}}$ ceases to exist above $T_c$ (Choupo <i>refuses</i> it, §1.2); and the species thermally <i>decompose</i> — the fixed-component assumption fails: do they still exist? |

The two edges — the ones you asked to watch — fail for opposite reasons but teach the same lesson. At the **cold edge** there is simply no data: every correlation is fitted to a window that starts well above  $73\text{ K}$ , the species are frozen, and a confident number there would be invention, not prediction. At the **hot edge** the theory keeps producing numbers — Antoine will gladly report a vapour pressure at  $800^\circ\text{C}$  — but the *physics* has moved on: past each  $T_c$  no saturation exists, and past a few hundred degrees the molecules are no longer the molecules in the **.dat**. In both cases the correct output is a *flag*, not a figure.

**Common pitfall.** No single model spans this axis. A run that quietly uses one  $\gamma$  model and one cubic from  $-200$  to  $1000^\circ\text{C}$  is not “general” — it is extrapolating through three regime changes in silence. The wide span is not one problem; it is five problems wearing one temperature axis.

**Intuition.** What Choupo ships *today* covers the middle of this span honestly: cubic EoS, ideal/NRTL/Wilson/UNIFAC activity, Antoine  $P^{\text{sat}}$ , Watson  $\Delta H_{\text{vap}}$ , three-phase VLLE by Gibbs minimisation, and azeotropic columns via the simultaneous MESH solver. It does *not* (yet) carry association models

(PC-SAFT/CPA) for the hydrogen-bonding non-ideality, nor electrolyte or wide-range near-critical packages. So for the H-bonding mid-span and the two edges, Choupo's job is not to fake accuracy it does not have — it is to predict where it can and to *say so* where it cannot. That honesty is the product.



## 2 The Choupo property architecture — the complete map

Everything in this guide hangs off one architecture. This chapter is the complete map, stated once; every later chapter is a room in this house.

### 2.1 The flow

`constant/thermoPhysPropDict` *declares* → the builder *loads, verifies and announces* (it never estimates) → the engine *computes* → the run log *confesses*.

Estimation is a curation-time act (the props bench writes a reviewable `.dat`); assembly-time refusal is loud (a missing declared parameter file, a  $k_{ij}$  regressed for another cubic, a vapour method the liquid path cannot serve). The log then shows the whole assembly: the VLE world, the reference rung of every group, and every pair parameter with its source file.

### 2.2 The grammar — the axes of a property package

| Axis                        | Slot                                                                    | What it is         |
|-----------------------------|-------------------------------------------------------------------------|--------------------|
| components                  | <code>components ( ... )</code>                                         | identity; a        |
| phases                      | <code>phases, phases/solid data</code>                                  | ions               |
| model per PHASE             | <code>equilibrium { formulation ...; liquid {...} vapour {...} }</code> | which pha          |
| convention per GROUP        | reference rungs ( <code>referenceBasis</code> )                         | ONE Gib            |
| correlation per COMPONENT   | the component <code>.dat</code> (+ overrides)                           | formulati          |
| parameter per PAIR          | <code>parameters { ...Pairs {...} }</code>                              | renormalis         |
| mixing per PROPERTY × PHASE | <code>mixing slots + effective{}</code>                                 | Raoult, so         |
| chemistry                   | <code>chemistry { salts (...); }</code>                                 | infinite dil       |
| package per UNIT            | <code>unit thermo { }</code>                                            | pure-speci         |
| phase equilibrium           | <code>phaseEquilibrium { }</code>                                       | pure corre         |
|                             |                                                                         | surface            |
|                             |                                                                         | declared F         |
|                             |                                                                         | to its sour        |
|                             |                                                                         | how pure           |
|                             |                                                                         | Nissan, var        |
|                             |                                                                         | closures (a        |
|                             |                                                                         | viscosity) l       |
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|                             |                                                                         | whole pack         |
|                             |                                                                         | single axes        |
|                             |                                                                         | in one pha         |
|                             |                                                                         | dited ( $\Delta H$ |
|                             |                                                                         | for each           |
|                             |                                                                         | imposed            |
|                             |                                                                         | DATUM              |
|                             |                                                                         | datum aqu          |
|                             |                                                                         | <pair-fil          |
|                             |                                                                         | surfaces,          |
|                             |                                                                         | implicit           |

Provenance is an *attribute of every value*, not an axis: each resolved number carries a quality word (`measured > fitted > correspondingStates > estimated > asserted`), and an `asserted` value — a pinned constant, a user polynomial, a user table — is refused without its written *why*.

### 2.3 The four VLE worlds

All selected by the *liquid* method slot; each validated against measured data in this guide:

| World                 | Liquid method                           | K-value                       | Natural home                 |
|-----------------------|-----------------------------------------|-------------------------------|------------------------------|
| $\gamma$ - $\varphi$  | activity.NRTL (Wilson, UNIQUAC, UNIFAC) | $\gamma_i P_i^{\text{sat}}/P$ | organic mixtures             |
| Henry                 | solution.henryDilute                    | $\gamma_i^* H_i/P$            | dissolved gases              |
| $\varphi$ - $\varphi$ | eos.SRK / eos.PengRobinson              | $\varphi_i^L/\varphi_i^V$     | high pressure, near-critical |
| electrolyte           | electrolyte.pitzer / .eNRTL             | ion activities, osmotic       | brines, salts, mixed solvent |

## 2.4 The five component routes

1. **Standard, by name** — `data/standards/components/` (exact-name  $O(1)$ , every value cited, the engine refuses to write there).
2. **Local, by name** — `data/local/components/` is the *private* working tier (gitignored; shipped empty). Records are runtime-usable, but every use is announced as `[local] UNVERIFIED`; a local record never shadows a standard one. Integrity and provenance gates make a local record inspectable, not automatically authoritative.
3. **New, defined in the case** — `myCase/constant/components/`: a new molecule (estimated and labelled), a new ion, a pseudo-component lump. The case runs self-contained from day one; promotion into the standards tree is a *requested, reviewed curation act* after validation.
4. **Standard + overlay** — a partial case file merges field-by-field over the standard entry (sample-specific measured data, a new model block); the run announces the merge.
5. **Solids** — solid reference rungs, solids on their own stream channel, and the iron rule: a dissolving salt's solid formation enthalpy is ION-DERIVED ( $H_f^{\text{solid}} = \sum \nu_i h_{f, \text{aq}, i} - \Delta H_{\text{soln}}$ ), never a stored duplicate.

Thus component resolution is **case-local overlay > standard > local (private, announced)**. This is an evidence ladder, not merely a search path: moving a file up the ladder requires review and validation, not a rename.

## 2.5 The ChemSep import route

The ChemSep 8.3 importer (`bin/curate/chemsep_to_choupo.py`) is deliberately reproducible and reviewable. It runs CAS-based de-duplication against the standard catalogue, then a structural/provenance audit and an internal  $P^{\text{sat}}(T_b)$  closure check on every imported component record — a closure check on the imported constants and predictive correlation, not an independent comparison with laboratory data.

The same import stages NRTL, UNIQUAC and Wilson binary-pair records, each passing identity, direction, licence, source-digest and exact unit-conversion checks. Their source does not provide a captured temperature-validity range, so they remain screening data and the runtime prints `[local] UNVERIFIED` when one is selected. Pair resolution is **case-local/inline > standard > local > announced ideal default**; local data can therefore extend a case without silently changing an approved standard.

The source boundary is intentional. The unmodified third-party database and its Artistic-2.0 licence are kept under `thirdParty/chemsep/` (gitignored); the normalised, attributed records are written to the *private data/local/* tier — they are never redistributed with the public repository. Literature values retain literature provenance; Ambrose–Walton and other closures retain predictive provenance. Missing  $C_p$ , formation, electrolyte, adsorption, kinetic or transport data are not inferred merely to make a record look complete.

The case additionally owns its *equipment physics*: reaction kinetics in `constant/reactions` (the heat of reaction always computed from the formation datum and announced), per-machine measured curves (a dryer's falling-rate data) in the unit's own folder, all inherited through the LOUD fractal cascade (plant → sector → unit).

## 2.6 The reference case

The whole architecture is exercised by one designed flagship, `tutorials/plant/lithiumBrinePlant` (under construction): salar brine through Pitzer ponds and crystallisers, NF membrane Mg/Li separation, solvent extraction with a student-defined extractant (the  $\gamma$ - $\varphi$  world and the audited Pitzer↔NRTL seam), electrodialysis, 60-bar CO<sub>2</sub> carbonation (the  $\varphi$ - $\varphi$  override and the Henry world), a radical-kinetics burner satellite for thermal NO<sub>x</sub>, batch recrystallisation recipes, a control satellite, and the full outer ring (fitting, sweeps, optimisation, sizing, costing, economics). Later chapters refer to its sectors by name.

## 2.7 The principle

No silent crutch, anywhere: every value is cited, estimated-and-labelled, overlaid-and-announced, or asserted-with-a-why — and the run log is where the whole assembly confesses. There is no ultimate model; hunting with the cat is honest engineering, hiding the cat is fraud.

## 3 The thermodynamic chain: how every equation here is derived

Every K-value, activity coefficient and Henry constant in this guide falls out of ONE chain of ideas. This section walks it once, so that no equation below ever has to be taken on faith. (Full derivations: the Theory Guide, chapters on flash, activity models, electrolytes and Henry's law.)

### 3.1 1. Equilibrium is a Gibbs minimum

The second law says an isolated system maximises entropy; recast at the conditions a process engineer actually controls — fixed  $T$  and  $P$  — it says a system minimises its **Gibbs energy**  $G$ . For a multicomponent, multiphase system,

$$dG = -S dT + V dP + \sum_i \mu_i dn_i, \quad \mu_i \equiv \left( \frac{\partial G}{\partial n_i} \right)_{T, P, n_{j \neq i}},$$

and at the minimum, moving a mole of  $i$  between phases must change nothing:

$$\boxed{\mu_i^\alpha = \mu_i^\beta} \quad \text{for every component, across every pair of phases.}$$

That single boxed statement is ALL the physics; everything else is bookkeeping to make it computable.

### 3.2 2. Fugacity: the workable currency

$\mu_i$  is awkward: its absolute value is unknowable and it dives to  $-\infty$  as  $x_i \rightarrow 0$ . Lewis (1901) defined the **fugacity**  $f_i$  through  $\mu_i = \mu_i^\circ + RT \ln(f_i/f_i^\circ)$  — an “effective pressure” that equals  $y_i P$  in an ideal gas. Because  $\ln$  is monotonic, the equilibrium condition becomes

$$\boxed{f_i^\alpha = f_i^\beta}$$

— equal fugacities, phase by phase, component by component. Every VLE “world” in Choupo is just a different, self-consistent way of WRITING  $f_i$  for a phase.

### 3.3 3. Two routes to a mixture fugacity

**Route A — an equation of state** (works for any phase the EoS can represent). Define the fugacity coefficient  $\varphi_i \equiv f_i/(y_i P)$ ; thermodynamics gives it exactly from the volumetric behaviour,

$$RT \ln \varphi_i = \int_V^\infty \left[ \left( \frac{\partial P}{\partial n_i} \right)_{T, V, n_j} - \frac{RT}{V} \right] dV - RT \ln Z,$$

so ONE model of  $P(T, V, \mathbf{n})$  — a cubic like SRK with its  $a(T)$ ,  $b$  and the pair correction  $k_{ij}$  — yields  $\varphi_i$  for BOTH the vapour root and the liquid root of the same cubic.

**Route B — an activity model** (liquids only). Split the liquid fugacity into a reference times a correction,  $f_i^L = \gamma_i x_i f_i^\circ$ , where the **activity coefficient**  $\gamma_i$  comes from a model of the excess Gibbs energy ( $G^E$ : NRTL, UNIQUAC, Wilson, Pitzer, eNRTL) via  $RT \ln \gamma_i = \partial G^E / \partial n_i$ . The price of Route B is choosing the **reference**  $f_i^\circ$  — and that choice is a CONVENTION, not physics:

| Convention            | Reference $f_i^\circ$                                                                            | Normalised so                                     |
|-----------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------|
| symmetric (Raoult)    | pure liquid: $P_i^{\text{sat}} (\times \varphi_i^{\text{sat}} \cdot \text{Poynting at high } P)$ | $\gamma_i \rightarrow 1$ as $x_i \rightarrow 1$   |
| unsymmetric (Henry)   | infinite dilution: $H_{i, \text{soln}} = \lim_{x \rightarrow 0} f_i/x_i$                         | $\gamma_i^* \rightarrow 1$ as $x_i \rightarrow 0$ |
| electrolyte (aqueous) | ion at infinite dilution, $H^+(\text{aq}) = 0$ datum                                             | $\gamma_\pm \rightarrow 1$ as $m \rightarrow 0$   |

A supercritical solute HAS no pure liquid, so its only definable anchor is the Henry limit — which is why dissolved  $\text{H}_2/\text{N}_2$  take the unsymmetric convention, never an Antoine curve extrapolated above  $T_c$ .

### 3.4 4. One Gibbs surface per phase — the legality rule

Gibbs–Duhem,  $\sum_i x_i d \ln \gamma_i = 0$  at constant  $T, P$ , ties every component's  $\gamma$  in a phase to every other's: they must all derive from ONE  $G^E$  (or one EoS) surface. Hence the rule that governs Choupo's **thermoPhysPropDict** grammar: *models per phase; conventions per component-group within the phase; correlations per component; parameters per pair*. Mixing two  $\gamma$ -models (or two cubics) in one phase violates Gibbs–Duhem and is refused. Mixing CONVENTIONS in one phase is legal — solvent symmetric, solutes unsymmetric — because a convention only renormalises the same surface.

### 3.5 5. The four worlds' K-values, assembled

Setting  $f_i^L = f_i^V$  with each route/convention gives every K-value this guide uses:

| world                 | $f_i^L$                                                                                                                           | $f_i^V$             | $K_i = y_i/x_i$                                  |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------|
| $\gamma$ - $\varphi$  | $\gamma_i x_i P_i^{\text{sat}}$                                                                                                   | $y_i \varphi_i P$   | $\frac{\gamma_i P_i^{\text{sat}}}{\varphi_i P}$  |
| Henry (KK/KI)         | $\gamma_i^* x_i H_i(T) e^{\bar{v}_i^\infty (P-P_s)/RT}$                                                                           | $y_i \varphi_i P$   | $\frac{\gamma_i^* H_i \mathcal{P}}{\varphi_i P}$ |
| $\varphi$ - $\varphi$ | $x_i \varphi_i^L P$                                                                                                               | $y_i \varphi_i^V P$ | $\frac{\varphi_i^L}{\varphi_i^V}$                |
| electrolyte           | ion activities from Pitzer/eNRTL on the aqueous rung;<br>full speciation (mass action + charge balance) where the case demands it |                     |                                                  |

Each row is one `equilibrium.formulation` (`gammaPhi`, `diluteSolution`, `phiPhi`, `electrolyteGammaPhi`); the run log announces which world is live. All four are validated in the tutorials against measured data — van Ness-consistent VLE sets ( $\gamma$ - $\varphi$ ), Wiebe/Larson solubilities (Henry,  $\sim 2\%$ ), the Stryjek N<sub>2</sub>-CH<sub>4</sub> isotherm ( $\varphi$ - $\varphi$ ,  $\sim 2\%$ ), and the Hamer–Wu / calorimetric electrolyte sets.

**Intuition.** The whole edifice is:  $G$  minimum  $\rightarrow$  equal  $\mu \rightarrow$  equal  $f \rightarrow$  pick HOW to write  $f$  per phase (an EoS or a  $G^E$  model + a reference convention)  $\rightarrow$  a K-value. When an equation in this guide looks arbitrary, walk it back up this chain — it never is.

## 4 Why a separate guide

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The Theory Guide answers *how a model works* — how a flash splits, how an EoS predicts  $Z$ , how Levenberg–Marquardt minimises a residual. This guide answers a question that comes *before* that: *where do the numbers that feed the models come from?* A simulation is only as good as the  $T_c$ , the NRTL pair, the heat of formation it stands on, and those are not handed down — they are estimated, fitted, and tested.

Choupo treats that as a first-class phase with its own binary (`choupoProps`) and its own lifecycle:

**CREATE** a component (from groups) → **CURATE** the mixture models (fit pairs, test data, choose by evidence) → **WIRE** the flowsheet → **SIMULATE**.

A `propsDict` of `operations` drives the first two stages. A case may have *no* `flowsheetDict` at all (a pure properties case — the compound bench), or carry one and use props to curate the thermo it will simulate. Same files, same engine; only the centre of the GUI differs.

**Intuition.** The usual reflex is to click a property method and trust a badge. Choupo’s stance (the “see, then decide” principle) is the opposite: you make the estimate or the fit *visible* — the group build-up, the model curve against the lab points, the consistency residual — and decide from the evidence in front of you. Estimation has error; the point is to see it.

## 5 Where curated data lives, and how a case selects it

You estimated a  $T_c$ , fitted an NRTL pair, tested a heat of solution. Two questions follow: *where does that number land*, and *how does a case use it*? Choupo’s answer is *declarative property packages with OpenFOAM-like explicit files*.

A curated value lands in a glass-box `.dat` in a tree organised by *what kind of thing* it is — the intrinsic property of one substance, the parameter of a pair, the equilibrium of a reaction:

```
data/standards/
  components/<name>.dat      intrinsic, one substance (Tc, Antoine, Cp)
                             + dissociatesTo { Na 1; Cl 1; } [salts]
  species/<name>.dat         one modelSpecies file per species
                             (Na, Cl, CO2aq, O2): formula + charge +
                             medium-tagged thermo; never fed to a flowsheet
  phases/solid/<phase>.dat   crystal phase (rho_p, k_v): halite, sylvite
  chemistry/.../<name>.dat   an equilibrium: dissolution, speciation
  parameters/.../<i>-<j>.dat  a pair: NRTL, Pitzer, eNRTL
  methods/<name>.dat         a model declaration (+ its reference basis)
  (the system is declared INLINE in the case's constant/thermoPhysPropDict
   -- THE CENTRE; there is no shared package catalogue)
```

The arity rule decides *which* file: a number that “stands alone with the molecule” (the  $T_c$  of water) lives with the component; a number that needs two names (an NRTL pair) cannot, and goes to `parameters/`. The Developer Guide (*the thermophysical property architecture*) gives the full layout and the eight data kinds.

**Declaring the system.** A case does not list loose models and parameters. It *declares* its thermophysical system — one record that names the components, the formulation, and the parameter files the run needs (active chemistry lives beside it in `constant/chemistryDict`):

```
recordType    thermoPhysicalPropertySystem;
schemaVersion 2;

components ( water NaCl );           // NaCl's dissociatesTo names the ions

equilibrium
{
    formulation electrolyteGammaPhi;

    aqueous
    {
        solvent          water;
        apparentComponents ( NaCl ); // stream carries the salt; the model
        activityModel { model Pitzer; } // activates its ions
        compositionBasis molality;
    }

    vapour { fugacityModel idealGas; }
}
% Pitzer pairs resolve by (cation,anion) name from parameters/; declared
% binaryParameters/binaryInteractions files load from their 'source' paths
% and refuse when missing.
```

The flow is one sentence: **thermoPhysPropDict declares, ThermoPackageBuilder assembles, ThermoPackage computes**. The declaration is the recipe; the builder loads what it names and mounts the runtime object the unit operations consume. An ordinary molecular case is the same record degenerate — no dissociation, no chemistry, a `gammaPhi` block with an NRTL or an EoS slot.

**Intuition.** The boundary you must respect: *curation estimates, runtime does not*. You may estimate, fit, and derive at the props bench — and you *see* it (§6, §7). Once a case runs, the builder only *loads* existing case, standard or local records; it never invents a missing property. For activity models, an unresolved binary

pair may use the documented ideal default, but that fallback is announced and appears in the provenance. A required record with no defined fallback stops the run with a clear message. So: estimate, look, promote — and inspect every local or defaulted rung used by the package.

## 6 Creating a component: Joback group contribution

When a component is missing from the catalogue and from the literature, you can *estimate* its pure-component constants from its molecular structure — the groups it is built from. Choupo implements the **Joback** method (Joback & Reid, 1987; tabulated in Poling, Prausnitz & O’Connell, *The Properties of Gases and Liquids*, 5th ed.), the `estimateComponent` property operation.

### 6.1 The method

Each first-order group contributes a tabulated increment. Summing the increments and substituting into Joback’s correlations gives the constants (with  $n_A$  the total number of atoms):

$$T_b = 198.2 + \sum_i n_i \Delta T_{b,i} \quad (4)$$

$$T_c = T_b / (0.584 + 0.965 \Sigma \Delta T_c - (\Sigma \Delta T_c)^2) \quad (5)$$

$$P_c = (0.113 + 0.0032 n_A - \Sigma \Delta P_c)^{-2} \quad [\text{bar}] \quad (6)$$

$$V_c = 17.5 + \sum_i n_i \Delta V_{c,i} \quad [\text{cm}^3/\text{mol}] \quad (7)$$

$$\Delta H_f^\circ(298) = 68.29 + \sum_i n_i \Delta H_{f,i} \quad [\text{kJ/mol, ideal gas}] \quad (8)$$

$$\Delta H_{\text{vap}}(T_b) = 15.30 + \sum_i n_i \Delta H_{v,i} \quad (9)$$

$$C_p^{\text{ig}}(T) = (\Sigma a - 37.93) + (\Sigma b + 0.210)T + (\Sigma c - 3.91 \times 10^{-4})T^2 + (\Sigma d + 2.06 \times 10^{-7})T^3. \quad (10)$$

The acentric factor  $\omega$  is *not* a group sum; Choupo derives it from  $(T_b, T_c, P_c)$  by the Lee–Kesler correlation, with  $T_{br} = T_b/T_c$  and  $P_c$  in atm:

$$\omega = \frac{-\ln P_c - 5.97214 + 6.09648/T_{br} + 1.28862 \ln T_{br} - 0.169347 T_{br}^6}{15.2518 - 15.6875/T_{br} - 13.4721 \ln T_{br} + 0.43577 T_{br}^6}. \quad (11)$$

A few first-order group increments (Joback & Reid; the full table is in the code, `src/propertyOps/EstimateComponent.cpp`):

| group            | $\Delta T_b$ | $\Delta T_c$ | $\Delta P_c$ | $\Delta H_f$ (kJ/mol) |
|------------------|--------------|--------------|--------------|-----------------------|
| –CH <sub>3</sub> | 23.58        | 0.0141       | –0.0012      | –76.45                |
| >CH <sub>2</sub> | 22.88        | 0.0189       | 0.0000       | –20.64                |
| –OH (alcohol)    | 92.88        | 0.0741       | 0.0112       | –208.04               |
| >C=O (ketone)    | 76.75        | 0.0380       | 0.0031       | –133.22               |
| aromatic =CH–    | 26.73        | 0.0082       | 0.0011       | 2.09                  |

### 6.2 The structure is identity: jobackGroups and bin/estimate

The molecular structure belongs *with the substance*, like the formula: a component’s `.dat` declares its groups once,

```
jobackGroups { CH3 2; ketone 1; }      // acetone = 2 x CH3 + 1 x >C=O
```

(the sibling of the UNIFAC `groups{}` block — a different table, the same idea), and estimation needs no case at all:

```
bin/estimate acetone      # case-local first, then the standard catalogue
bin/estimate path/to/new.dat  # or an explicit file
```

The command runs the Joback engine on the component’s own groups, prints the full glass-box build-up (each group’s increment, the closure chain, deviations against any **reference** block), and drops *one* stable proposal — `<name>.estimated.dat` — next to the source, for you to review and promote by hand. It never writes into the frozen `data/standards/` tree, and the solver still refuses to estimate at run time: estimation is a *curation* act with a paper trail, not a silent fill-in.

The `estimateComponent` operation remains the pedagogical form — declare the groups inline in a `propsDict` when you want the estimate itself to be the lesson, as in the worked example below.



### 6.3 Worked example: acetone

Acetone,  $\text{CH}_3\text{—CO—CH}_3$ , decomposes into  $2 \times \text{CH}_3$  and  $1 \times \text{ketone}$ . The operation prints the build-up and validates against a **reference** block:

| Property           | Joback                     | Reference (NIST)         | Dev.  |
|--------------------|----------------------------|--------------------------|-------|
| $T_b$              | 322.1 K                    | 329.2 K                  | −2.2% |
| $T_c$              | 500.6 K                    | 508.1 K                  | −1.5% |
| $P_c$              | 48.0 bar                   | 47.0 bar                 | +2.2% |
| $V_c$              | 209.5 cm <sup>3</sup> /mol | 209 cm <sup>3</sup> /mol | ≈ 0   |
| $\omega$           | 0.295                      | 0.307                    | −3.9% |
| $\Delta H_f^\circ$ | −217.8 kJ/mol              | −217.1 kJ/mol            | +0.3% |

Within a few percent on the criticals and almost exact on  $\Delta H_f$  — and you *see* the error, because the reference sits beside the estimate.

**Intuition. Know the method's limits.** Joback is weak on  $T_b$  of strongly hydrogen-bonding species (ethanol's  $T_b$  comes out ~14 K low), and it yields *no* Antoine / vapour-pressure parameters — those need a fit (§7) or a corresponding-states correlation, a separate step. An estimate is a starting point you then anchor to data, never a final answer.

**Runnable case.** `tutorials/props/estimate/estimate_acetone` — the example above, groups in, validated estimate out.

## 7 Curating mixture models: fitting binary pairs

A pure-component set is not enough for a mixture: the non-ideality lives in the *binary interactions* (NRTL, Wilson). Those parameters are regressed against experimental VLE data by Levenberg–Marquardt — the `fitParameters` operation.

### 7.1 The activity-coefficient models

For a binary, the **NRTL** model gives the liquid-phase activity coefficients from three adjustable numbers per pair —  $\tau_{12}, \tau_{21}$  (temperature-dependent) and the non-randomness  $\alpha$ :

$$\tau_{ij} = a_{ij} + \frac{b_{ij}}{T}, \quad G_{ij} = \exp(-\alpha \tau_{ij}), \quad (12)$$

$$\ln \gamma_1 = x_2^2 \left[ \tau_{21} \left( \frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right], \quad (13)$$

and  $\ln \gamma_2$  by swapping the indices. So a pair carries four fittable coefficients ( $a_{12}, b_{12}, a_{21}, b_{21}$ ) plus a fixed  $\alpha$  (often 0.3). **Wilson** is the two-parameter alternative,

$$\ln \gamma_1 = -\ln(x_1 + \Lambda_{12}x_2) + x_2 \left[ \frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1} \right], \quad (14)$$

which cannot represent a liquid–liquid split (use NRTL when one is suspected).

### 7.2 Predicting without pairs: COSMO-SAC

NRTL and Wilson need a fitted pair — somebody measured that binary. **COSMO-SAC** is the predictive alternative: the activity coefficients come from each compound’s *sigma profile* — the histogram of screening charge density over its molecular surface, from a quantum-chemical calculation done *once, elsewhere, at curation time* — and no binary parameter exists at all. Choupo implements the 2002 variant (Lin & Sandler), following the NIST benchmark implementation, as a plain activity model: same interface as NRTL, selected the same way.

A compound carries its profile in its own `.dat`, as one or more *named parameter sets* (a compound may carry profiles from several public collections, each labelled with its source):

```
cosmo
{
  VT2005          // set name = its source collection
  {
    variant       cosmoSAC2002;
    source        "Mullins et al., IECR 45 (2006) 4389";
    area          [-] 106.3;      // A^2
    volume        [-] 118.8;      // A^3
    sigmaProfile  ( ... );        // 51 points, -0.025..0.025 e/A^2
  }
}
```

The case selects the model and, when a compound carries more than one set, names which one:

```
activityModel { model cosmoSAC; source VT2005; }
```

A compound with a single set needs no `source` (its lone set is the default); a multi-set compound without one is a loud error, as is a missing `cosmo` block or a set whose declared `variant` does not match the implemented constants — a profile is never silently combined with another variant’s parameters. The standard catalogue ships VT2005 sets for 77 components. Validation: a pure component gives  $\ln \gamma = 0$  exactly, and the implementation is cross-checked bit-for-bit against an independent code on the same equations and profiles. What Choupo deliberately does *not* do is generate profiles (that is quantum chemistry, a curation act outside the simulator) or bulk-import thousands of compounds. Reference case: `tutorials/props/molecular/cosmoSAC01_water_ethanol`.

### 7.3 The residual the fit minimises

The measurable is the bubble temperature. At a fixed pressure  $P$  and liquid composition  $x$ , the bubble point is the  $T$  that closes modified Raoult,

$$\sum_i x_i \gamma_i(x, T) P_i^{\text{sat}}(T) = P \quad (15)$$

(a Newton-1D in  $T$ ;  $P_i^{\text{sat}}$  from Antoine). Levenberg–Marquardt then drives the pair coefficients  $\theta = (a_{12}, b_{12}, a_{21}, b_{21})$  to minimise the sum of squared bubble-temperature residuals against the data,

$$\min_{\theta} \sum_k [T_{\text{bub}}(x_k; \theta) - T_k^{\text{exp}}]^2. \quad (16)$$

**Intuition.** A low residual is not a good fit. At a *single* pressure  $\tau_{ij} = a_{ij} + b_{ij}/T$  is nearly linear in both  $a_{ij}$  and  $b_{ij}/T$  over the narrow  $T$  range of one isotherm/isobar, so  $a$  and  $b$  are *correlated* — many  $(a, b)$  pairs fit equally well and the optimum is a valley, not a point. `fitParameters` forms  $J^T J$  at the optimum and reports the parameter correlation + confidence intervals so you SEE this; the Theory Guide’s *Levenberg–Marquardt* and *What is fittable* chapters derive the diagnostic. Fit across pressures, or fix  $b$ , to break it.

## 7.4 See, then decide: the model overlay

Before fitting, look. Declare one scan `operation` per candidate model (ideal, NRTL, Wilson), emitting the  $(x, T_{\text{bubble}}, y_{\text{eq}})$  trio, and an `experimental` `{}` entry that overlays them — and, with a `dataset`, draws the measured points on top. The student sees which model the data actually picks: for ethanol/water at 1 atm the ideal curve misses the azeotrope entirely while NRTL lands on it.

**Runnable case.** `tutorials/props/compare/compare_vle_etoh_water` — ideal vs. NRTL bubble/dew curves overlaid with measured VLE points (Carey & Lewis, 1932).

## 7.5 Fit, then promote

When a model earns its keep, `fitParameters` regresses its coefficients to the data and writes a proposal to `constant/parameters/<model>/<pair>.fit-<date>.dat`. Promotion is a deliberate, visible act: you inspect the parity plot and the identifiability statistics (confidence intervals, parameter correlation) the fit reports — never a “good fit” badge — and copy the proposal into the case’s `constant/` when you trust it.

## 8 Testing the data itself: consistency

Before fitting a model *to* data, test whether the data is even thermodynamically consistent — a fit to bad data is a polished wrong answer. The `vleConsistency` operation does this in three steps.

### 8.1 Step 1 — activity coefficients straight from the data

It does not assume a model. From each measured  $(x_1, y_1, T)$  at the run pressure  $P$ , modified Raoult gives the activity coefficient of each component directly:

$$\gamma_i = \frac{y_i P}{x_i P_i^{\text{sat}}(T)}, \quad P_i^{\text{sat}}(T) \text{ from Antoine.} \quad (17)$$

These are *experimental*  $\gamma_i$  — the test now asks whether they obey thermodynamics.

### 8.2 Step 2 — the Gibbs–Duhem point test

The Gibbs–Duhem equation is the constraint every real mixture must satisfy at constant  $T, P$ :

$$\sum_i x_i d \ln \gamma_i = 0 \implies x_1 \frac{d \ln \gamma_1}{dx_1} + x_2 \frac{d \ln \gamma_2}{dx_1} = 0 \quad (\text{binary}). \quad (18)$$

The operation evaluates the left-hand side at each interior point (central differences in  $x_1$ ) and reports its max and mean magnitude — the *point residual*. A consistent set keeps it near zero. (Isobaric data adds a small  $-H^E/RT^2 \cdot dT/dx$  term, dropped here; stated as a caveat, not hidden.)

### 8.3 Step 3 — the Herington area test

Integrate (18) across the whole composition range: for an isothermal set Gibbs–Duhem forces the net area of  $\ln(\gamma_1/\gamma_2)$  to vanish. Herington’s isobaric form compares that net area to the absolute area and corrects for the boiling-range:

$$D = 100 \frac{\left| \int_0^1 \ln \frac{\gamma_1}{\gamma_2} dx_1 \right|}{\int_0^1 \left| \ln \frac{\gamma_1}{\gamma_2} \right| dx_1}, \quad J = 150 \frac{T_{\text{max}} - T_{\text{min}}}{T_{\text{min}}}, \quad (19)$$

(both integrals by the trapezoid rule over the sorted points). The set **passes** when  $|D - J| < 10$ . The GUI’s consistency card plots  $\ln \gamma_1$ ,  $\ln \gamma_2$  and  $\ln(\gamma_1/\gamma_2)$  vs  $x_1$  with the net area shaded — you SEE the  $\pm$  areas cancel, not just the number.

Finally the infinite-dilution coefficients  $\gamma_1^\infty, \gamma_2^\infty$  (the dilute-end limits) are reported — the most stringent point of any model later fitted to the set.

**Runnable case.** `tutorials/props/compare/compare_vle_etoh_water` — the `vleConsistency` op on the Carey–Lewis ethanol/water data (Herington  $|D - J| = 4.6 < 10$ , PASS).

This guide grows with the props pillar. Planned: a second group-contribution method (Constantinou–Gani) to compare against Joback; group contribution for liquid viscosity; a structure (SMILES) front-end for the compound bench.